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Viral genome sequence datasets display pervasive evidence of strand-specific substitution biases that are best described using non-reversible nucleotide substitution models

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Abstract

Background

The vast majority of phylogenetic trees are inferred from molecular sequence data (nucleotides or amino acids) using time-reversible evolutionary models which assume that, for any pair of nucleotide or amino acid characters, the relative rate of X to Y substitution is the same as the relative rate of Y to X substitution. However, this reversibility assumption is unlikely to accurately reflect the actual underlying biochemical and/or evolutionary processes that lead to the fixation of substitutions. Here, we use empirical viral genome sequence data to reveal that evolutionary non-reversibility is pervasive among most groups of viruses. Specifically, we consider two non-reversible nucleotide substitution models: (1) a 6-rate non-reversible model (NREV6) in which Watson-Crick complementary substitutions occur at identical relative rates and which might therefor be most applicable to analyzing the evolution of genomes where both complementary strands are subject to the same mutational processes (such as might be expected for double-stranded (ds) RNA or dsDNA genomes); and (2) a 12-rate non-reversible model (NREV12) in which all relative substitution types are free to occur at different rates and which might therefore be applicable to analyzing the evolution of genomes where the complementary genome strands are subject to different mutational processes (such as might be expected for viruses with single-stranded (ss) RNA or ssDNA genomes).

Results

Using likelihood ratio and Akaike Information Criterion-based model tests, we show that, surprisingly, NREV12 provided a significantly better fit to 21/31 dsRNA and 20/30 dsDNA datasets than did the general time reversible (GTR) and NREV6 models with NREV6 providing a better fit than NREV12 and GTR in only 5/30 dsDNA and 2/31 dsRNA datasets. As expected, NREV12 provided a significantly better fit to 24/33 ssDNA and 40/47 ssRNA datasets. Next, we used simulations to show that increasing degrees of strand-specific

substitution bias decrease the accuracy of phylogenetic inference irrespective of whether GTR or NREV12 is used to describe mutational processes. However, in cases where strand-specific substitution biases are extreme (such as in SARS-CoV-2 and Torque teno sus virus datasets) NREV12 tends to yield more accurate phylogenetic trees than those obtained using GTR.

Conclusion

We show that NREV12 should, be seriously considered during the model selection phase of phylogenetic analyses involving viral genomic sequences.

eLife assessment

This **valuable** study revisits the effects of substitution model selection on phylogenetics by comparing reversible and non-reversible DNA substitution models. The authors provide evidence that 1) non time-reversible models sometimes perform better than general time-reversible models when inferring phylogenetic trees out of simulated viral genome sequence data sets, and that 2) non time-reversible models can fit the real data better than the reversible substitution models commonly used in phylogenetics, a finding consistent with previous work. However, the methods are **incomplete** in supporting the main conclusion of the manuscript, that is that non time-reversible models should be incorporated in the model selection process for these data sets.

Background

Modelling the nucleotide substitution processes that underly the diversification of virus genome sequences is at the heart of many viral evolutionary analyses. The most widely used nucleotide substitution models belong to the general time reversible (GTR) family (Tavaré, 1986) that assume that the Markov process of evolution will occur in the same way both forward and backward in time such that, when the arrow of time is inverted, the forward process cannot be distinguished from the backward process (Hoff, Orf, Riehm, Darriba, & Stamatakis, 2016), (Lio & Goldman, 1998), (Tavaré, 1986).

The essence of the GTR model is captured in the definition of its instantaneous rate matrix in Eq. (1); a matrix that models the rates at which the four different nucleotides {A,C,G,T} are exchanged, ensuring that the detailed reversibility balance condition: $Q_{ji}\,\pi_i = Q_{ij}\,\pi_j$ (where Q_{ji} is the instantaneous rate of change from j to i and π_i is the equilibrium probability of state i) is met (Squartini & Arndt, 2008) (Posada, 2003). The instantaneous rate matrix of the GTR model consists of six parameters (a, b, c, d, e, and f) which indicate the relative rates from base i to j in the state space {A, C, G, T} and the π_i's are the equilibrium frequencies of each base

$$Q = \{q_{ij}\} = \begin{pmatrix} - & a\pi_A & b\pi_A & d\pi_A \\ a\pi_C & - & c\pi_C & e\pi_C \\ b\pi_G & c\pi_G & - & f\pi_G \\ d\pi_T & e\pi_T & f\pi_T & - \end{pmatrix}$$

The rate matrix in Eq. (1) is symmetrical in that, for example, the relative rate at which A changes to G is the same as the relative rate at which G changes to A.

Time reversible nucleotide substitution models such as GTR form the basis of almost all nucleotide sequence-focused evolutionary analyses (including those involving eukaryotes, prokaryotes, and viruses) (Lefort, Longueville, & Gascuel, 2017), (Posada & Crandall, 2001), (Minin, Abdo, Joyce, & Sullivan, 2003).

The reliability of a phylogenetic tree constructed using a particular nucleotide sequence dataset should be maximized when the evolutionary models used to construct the tree accurately reflect the evolutionary processes that yielded the nucleotide sequence dataset (Buckley & Cunningham, 2002), (Ripplinger & Sullivan, 2008), (Hoff, Orf, Riehm, Darriba, & Stamatakis, 2016). The suitability of different models for describing the evolution of DNA or RNA sequences is, therefore, expected to depend to some degree on the biological and environmental contexts of the sequences being analyzed.

Mutations in viral genomes arise due to diverse biotic (such as replication enzyme infidelities, RNA/DNA editing enzymes) and abiotic (such as ionizing radiation, inorganic oxidizers and chemical mutagens) factors (Sanjuán & Domingo-Calap, 2016). Mutagenic chemical reactions or types of radiation that, for example, cause G to A or C to U mutations in DNA or RNA, are not the same as those that cause A to G or U to C mutations (Cheng, Cahill, Kasai, Nishimura, & Loeb, 1992), (Nguyen, et al., 1992), (Chelico, Pham, Calabrese, & Goodman, 2006), (Sharma, Patnaik, Taggart, & Baysal, 20016). It should not be expected, therefore, that the relative rates of G to A substitution will equal as the relative rates of A to G substitution. Instead, in evolving double-stranded (ds) DNA and dsRNA molecules where both strands of the genome are in existence for similar amounts of time, both G to A and C to T substitutions should occur at relatively similar rates. Therefore, for nucleotide sequence datasets derived from any organisms with dsDNA or dsRNA genomes, a non-reversible nucleotide substitution model with a different relative substitution rate category for each of the six possible pairs of complementary nucleotide substitutions (e.g. NREV6 in Eq. (2)), might plausibly provide a better description of mutational processes than GTR (Baele, Van de Peer, & Vansteelandt, 2010), (Wickner, 1993).

$$Q = \{q_{ij}\} = \begin{pmatrix} - & a\pi_A & b\pi_A & c\pi_A \\ f\pi_C & - & d\pi_C & e\pi_C \\ e\pi_G & d\pi_G & - & f\pi_G \\ c\pi_T & b\pi_T & a\pi_T & - \end{pmatrix}$$

In the case of single-stranded (ss) RNA viruses, ssDNA viruses, retroviruses, and dsRNA/dsDNA viruses where the two complementary genome strands do not exist for equal amounts of time (Yu, et al., 2004), a model where all twelve different substitutions are free to occur at different rates might be best. Specifically, with ssRNA viruses, ssDNA viruses, and retroviruses, only one of the genome strands (called the virion strand) is packaged into viral particles for transmission and, in many dsRNA viruses, the genome strand that is translated into proteins (called the + strand) exists for longer during the life cycle than does the complementary (or −) strand (Bruslind, 2020), (Onwubiko, et al., 2020). In all these viruses, some degree of strand-specific substitution bias is expected to occur (Van Der Walt, Martin, Varsani, Polston, & Rybicki, 2008), (Polak & Arndt, 2008) such that NREV6 might be anticipated to provide a poorer description of mutational processes than a model such as NREV12 (Eq. (3)) where each of the twelve different types of substitution has a different relative rate (Baele, Van de Peer, & Vansteelandt, 2010).

$$Q = \{q_{ij}\} = \begin{pmatrix} - & a\pi_A & b\pi_A & c\pi_A \\ g\pi_C & - & d\pi_C & e\pi_C \\ h\pi_G & i\pi_G & - & f\pi_G \\ j\pi_T & k\pi_T & l\pi_T & - \end{pmatrix}$$

Because non-reversible models consider the directionality of evolution, they could, in some cases, be used to identify root nodes of phylogenetic trees (Yap & Speed, 2005), (Boussau & Gouy, 2006). It is, however, unclear whether non-reversible models might, in certain situations at least, perform better than reversible models in the context of phylogenetic inference. Although it is possible to use non-reversible nucleotide substitution models such as NREV6 and NREV12 during maximum likelihood-based phylogenetic inference with computer programs such as IQ-TREE (Nguyen, Schmidt, Von Haeseler, & Minh, 2015), these models are still not routinely used for phylogenetic inference. This is in part because non-reversible models render several commonly used algorithmic techniques for efficient likelihood computation inapplicable. It is also in part because it remains undetermined whether, under conditions where strand-specific substitution biases are evident, non-reversible models consistently yield substantially more accurate phylogenetic trees than reversible models.

Here we present evidence that strand-specific nucleotide substitution biases are common within virus genomic sequence datasets such that NREV12 generally provides a significantly better fit than both GTR and NREV6 for such datasets. We then use simulations to demonstrate that whereas strand-specific nucleotide substitution biases reduce the accuracy of phylogenetic inference irrespective of whether GTR or NREV12 are used, when these biases become extreme use of NREV12 can yield significantly more accurate phylogenetic trees than GTR.

Materials And Methods

Virus sequence datasets and phylogenetic trees

We obtained nucleotide sequences from the National Centre for Biotechnology Information Taxonomy database (http://www.ncbi.nlm.nih.gov/taxonomy) and the Los Alamos National Laboratory HIV sequence database (https://www.hiv.lanl.gov/content/index). These included gene and whole-genome sequences for viruses with ssRNA, ssDNA, dsRNA, and dsDNA genomes (datasets are summarized in supplementary Table S1). An outgroup sequence from a closely related virus species was added to each dataset to help root phylogenetic trees accurately. The sequences in each of the datasets were aligned using MUSCLE (Edgar, 2004) implemented in Aliview (Larsson, 2014), and maximum likelihood phylogenetic trees were constructed from each alignment using RAxML v8.2 (Stamatakis, 2016).

Model Testing

We evaluated the fit of NREV12, NREV6, and GTR to the 141 individual sequence datasets using a previously published model test (Harkins, et al., 2009) formulated in the HyPhy scripting language (Pond, Frost, & Muse, 2005). This script (which can be obtained from https://github.com/veg/hyphy-analyses/tree/master/NucleotideNonREV) took as input a rooted maximum likelihood phylogenetic tree (minus the rooting sequence) and its corresponding nucleotide sequence alignment. The first step of the model testing process involved the harvesting of nucleotide sequences

$$\left(\frac{\pi_{\mathrm{i}}}{\sum_{\mathrm{i}=1}^{4}\pi_{\mathrm{i}}}\right)$$

from the sequence alignment into a vector of frequencies $\pi = \{\pi_A, \pi_C, \pi_G, \pi_T\}$ called the frequency distribution matrix containing three free parameters with that of nucleotide T conditioned at absolute $(1 - \pi_A - \pi_C - \pi_G)$.

Once the frequencies were harvested, the first stochastic rate matrix consisting only of the relative rates picked from a gamma distribution was defined to satisfy the reversibility conditions of relative rates being equal in reverse i.e., rAG = rGA, rAC = rCA, rAT = rTA, rCG = rGC, rTG = rGT, rCT = rTC. Thereafter the instantaneous rate matrix Q was calculated by multiplying the relative substitution rates by the appropriate nucleotide frequencies which were used to form the GTR model probability transition matrix, P by $P(t) = e^{Qt}$. The role of the GTR model during model testing was to model mutations along the branches of tree, T. Given parameters of the GTR model describing (1) equilibrium nucleotide frequencies, and (2) the nucleotide substitution process, the likelihood, L, of the observed data, D, was calculated with the values of all the independent parameters, Θ, being investigated to find the combination that maximized the value of L (maximizing L (| D, T). The value of the log likelihood (lnL) under the GTR model was, at this point, stored for future comparisons with those of the NREV6 and NREV12 models.

The model was then changed to one satisfying the complementary relative reversibility conditions of the NREV6 model: i.e. rAG = rTC, rAC = rTG, rAT = rTA, rCG = rGC, rCA = rGT, rCT = rGA. The lnL of observing the data, D, given the tree, T, under the NREV6 model was calculated and stored for later comparisons. The model was then changed to the NREV12 model for which relative rates were defined such that it satisfied complete non-reversibility.

For each dataset we then used the lnL scores and numbers of free parameters in the three models for likelihood ratio tests (LRTs; (Anisimova, Bielawski, & Yang, 2001)) to determine whether (1) NREV12 fitted the data significantly better than GTR and (2) whether NREV12 fitted the data significantly better than NREV6. Specifically, for the NREV12 vs GTR comparison we calculated the LRTstatisticas2 $(\ln L_{NREV12} - \ln L_{GTR})$ with the p value being calculated as 1 − chi (LRT, $df_{NREV12} - df_{GTR}$). For the NREV12vsGTR comparison we calculated the LRTstatisticas2 $(\ln L_{NREV12} - \ln L_{NREV6})$ with the p value being calculated as 1 − chi (LRT, $df_{NREV12} - df_{NREV6}$).

Further, the lnL scores and numbers of free parameters for each model were used to calculate AIC scores for each of the models (Posada & Buckley, 2004) which enabled us to identify which of the three models fit the data best. The model with the lowest AIC score was selected as the best fitting model with the AIC scores for the different models being calculated as follows:

$$\begin{aligned}\mathrm{AIC_{GTR}} &= 2\left(\mathrm{df_{GTR}}\right) - 2(\mathrm{lnL_{GTR}})\\ \mathrm{AIC_{NREV6}} &= 2\left(\mathrm{df_{NREV6}}\right) - 2(\mathrm{lnL_{NREV6}}),\\ \mathrm{AIC_{NREV12}} &= 2\left(\mathrm{df_{NREV12}}\right) - 2(\mathrm{lnL_{NREV12}}).\end{aligned}$$

Quantification of non-reversibility.

We further defined the degree of non-reversibility (DNR) as the absolute difference between the relative rate differences of two nucleotide pairs; i.e. for two nucleotides, *x* and *y*, there exists a relative rate of *x* to *y* substitutions that we will refer to as *m*, and a relative rate *y* to *x* substitutions that we will refer to as n. Under the NREV12 model, the degree of non-reversibility (DNR) between x and y is defined simply as the absolute difference between m and n:($|m$-$n|$) and will hereby be referred to as ij_{DNR} were i and j are two nucleotides. We use DNR as a mathematical representation of the degree to which the rates of all pairs of reverse substitutions differ from one another. For each of 140 individual viral sequence datasets we calculated the average DNR of the six ij_{DNR} estimates inferred using the NREV12 model.

Simulations for testing the impact of non-reversible evolution on phylogenetic inference

We tested the accuracy of phylogenetic tree inference under reversible and non-reversible models using simulated datasets with varying average pairwise nucleotide sequence identities (APIs) evolved under the NREV12 model with different degrees of non-reversibility (DNR). The goal of these tests was not to exhaustively evaluate model misspecification issues during phylogenetic tree inference, but rather to check, in instances where viral taxa are known to be evolving in a detectably non-reversible manner (i.e. where NREV12 or NREV6 fits the data better than GTR), whether not accounting for this might decrease the accuracy of phylogenetic inference. Using IQTREE, a phylogenetic inference program that has the option to apply an NREV12-like model (referred to in IQTEE as the UNREST model), a phylogenetic tree was inferred from an alignment of real sequences (Avian Leukosis virus) with an average sequence identity (API) of ~ 90%. The branch lengths on this tree were then scaled to create four other phylogenetic trees representing sequences with approximately 95%, 85%, 80% and 75% API. These five trees are hereafter referred to as “true” trees and each individual tree was used as the starting point of a different set of simulations.

To test whether failure to account for non-reversibility might decrease the accuracy of phylogenetic inference we simulated the evolution of 5,500 nucleotide sequence alignments evolved non-reversibly under varying DNR along the five true phylogenetic trees: 100 datasets per true tree per simulated degree of non-reversibility (DNR). Specifically, simulations were done using HyPhy (Pond, Frost, & Muse, 2005) with relative rates ranging from a completely reversible matrix i.e., CA = AC = 0.166, GA = AG = 1, AT = TA = 0.14, GC = CG = 0.131, TG = GT = 0.188 and CT = TC = 1.101 – representing DNR = 0 – through matrices with DNR = 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20 (Table 1). These baseline simulated substitution rates are reflective of those seen in empirical viral nucleotide sequence datasets.

Table 1

Relative rate change for C to A, G to A, A to T, G to C, T to G and C to T mutations under the 11 degrees of non-reversibility alongside the maintained rates for A to C, A to G, T to A, C to G, G to T, and T to C

Degree of Non-Reversibility (DNR)	Relative rates of different nucleotide substitution types (from-to)											
	C-A	A-C	G-A	A-G	A-T	T-A	G-C	C-G	T-G	G-T	C-T	T-C
0	0.166	0.166	1	1	0.14	0.14	0.131	0.131	0.118	0.118	1.101	1.101
2	2.166	0.166	3	1	2.14	0.14	2.131	0.131	2.118	0.118	3.101	1.101
4	4.166	0.166	5	1	4.14	0.14	4.131	0.131	4.118	0.118	5.101	1.101
6	6.166	0.166	7	1	6.14	0.14	6.131	0.131	6.118	0.118	7.101	1.101
8	8.166	0.166	9	1	8.14	0.14	8.131	0.131	8.118	0.118	9.101	1.101
10	10.166	0.166	11	1	10.14	0.14	10.131	0.131	10.118	0.118	11.101	1.101
12	12.166	0.166	13	1	12.14	0.14	12.131	0.131	12.118	0.118	13.101	1.101
14	14.166	0.166	15	1	14.14	0.14	14.131	0.131	14.118	0.118	15.101	1.101
16	16.166	0.166	17	1	16.14	0.14	16.131	0.131	16.118	0.118	17.101	1.101
18	18.166	0.166	19	1	18.14	0.14	18.131	0.131	18.118	0.118	19.101	1.101
20	20.166	0.166	21	1	20.14	0.14	20.131	0.131	20.118	0.118	21.101	1.101

At each DNR, the relative rates used conformed to standard measures of non-reversibility under the Kolmogorov conditions according to which non-reversibly evolving sequence datasets should yield three irreversibility indices (IRI1, IRI,2 and IRI3) that are all non-zero (Squartini & Arndt, 2008). Accordingly, whereas a for simulated datasets where DNR was 0, the IRI1, IRI2 and IRI3 indices were all approximately zero indicating that the sequences in these datasets had (as expected), evolved in a time-reversible manner, for datasets where DNR was greater than 0, the IRI1, IRI12 and IRI3 indices were all different from zero indicating that the sequences in these datasets had indeed evolved in a time non-reversible manner. Further, it should be noted that all simulations under NREV12 were performed under the stationarity criterion:

$\pi e^{QT} = \pi$ (where Q is the rate matrix and π is the nucleotide frequency distribution and $t \geq 0$).

Quantifying the accuracy of phylogenetic inferences

We used the weighted Robinson-Foulds metric (wRF; implemented in the R phangorn package (Schliep, 2011)) to quantify differences between the true trees used to simulate datasets and the trees inferred from these datasets using the GTR or NREV12 models. wRF considers differences between both the topology and branch lengths of actual and inferred trees (Kuhner, 2015), (Robinson & Foulds, 1981).

Results And Discussion

Non-reversible nucleotide substitution models generally provide a better fit than reversible models to virus sequence datasets

We tested for evidence of non-reversibility of the nucleotide substitution process in 141 virus sequence datasets (33 ssDNA virus datasets, 30 dsDNA virus datasets, 31 dsRNA virus datasets, and 47 ssRNA virus datasets, all consisting of either full genome sequences (for unsegmented viruses), or complete genome component sequences (for viruses with segmented genomes). Specifically, for each dataset, we compared the goodness-of-fit of the GTR + G, NREV6 + G, and NREV12 + G models (where G represents gamma-distributed rates).

Given that dsDNA viruses such as adenoviruses, papillomaviruses and herpesviruses have both their DNA strands in existence for similar amounts of time before DNA-dependant-DNA polymerase enzymes copy both their + and − DNA strands during replication (Hanson, 2009), we had anticipated that the best fitting substitution model for sequence datasets of these viruses would be NREV6. Using weighted AIC scores to reveal trends of model support (Fig. 2), it is surprising that NREV12 was overall the best-supported model (illustrated by the redder hues around the top corner of the dsDNA plot in Fig. 2. Out of the 30 dsDNA datasets considered, we found that NREV6 provided the best fit to five datasets (HPV18, HPV45, HPV16, BPV, and SV40) and GTR provided the best fit to five (Alphapapilloma virus 6, JC polyomavirus, DPV, RTBV, and DBAV). NREV12 was the best fitting model for the remaining 20 datasets (Table 1). Further, LRTs revealed strong overall support for NREV12, with this model providing a significantly better fit ($p < 0.05$) than NREV6 for 25/30 of the dsDNA datasets and a significantly better fit than GTR for 24/30 of the datasets.

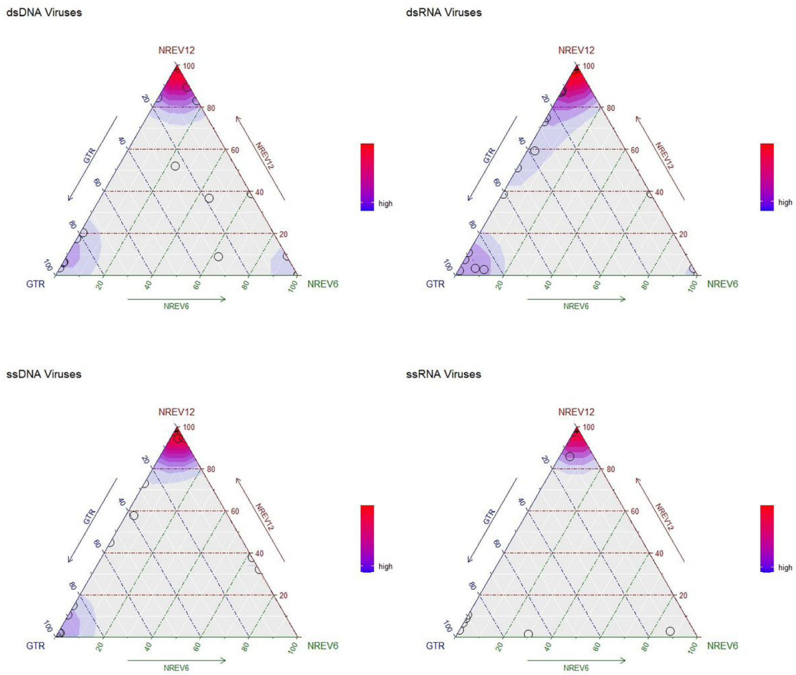

Figure 1

Ternary plots illustrating the relative fit of the NREV12, NREV6, and GTR nucleotide substitution models based on weighted AIC scores for 30 dsDNA, 31 dsRNA, 33 ssDNA, and 47 ssRNA virus nucleotide sequence datasets.

These plots were produced using the Akaike weights function with an overlaid density function (implemented in the qpcR package of RStudio (Ritz & Spiess, 2008) to indicate point densities. Each model is represented by a corner of the triangles, and each circle represents the relative fit of each of the three models to a single nucleotide sequence dataset. The sides of the triangle represent model support axes ranging from 0-100%, with the position of a circle in relation to each of the sides of the triangle indicating the probability of models best describing the nucleotide sequence dataset that is represented by that point. Whereas strong red colours represent a very high density of nucleotide sequence datasets that favour a particular model, bluer colours indicate a lower, but still substantial, density of datasets that favour a model.

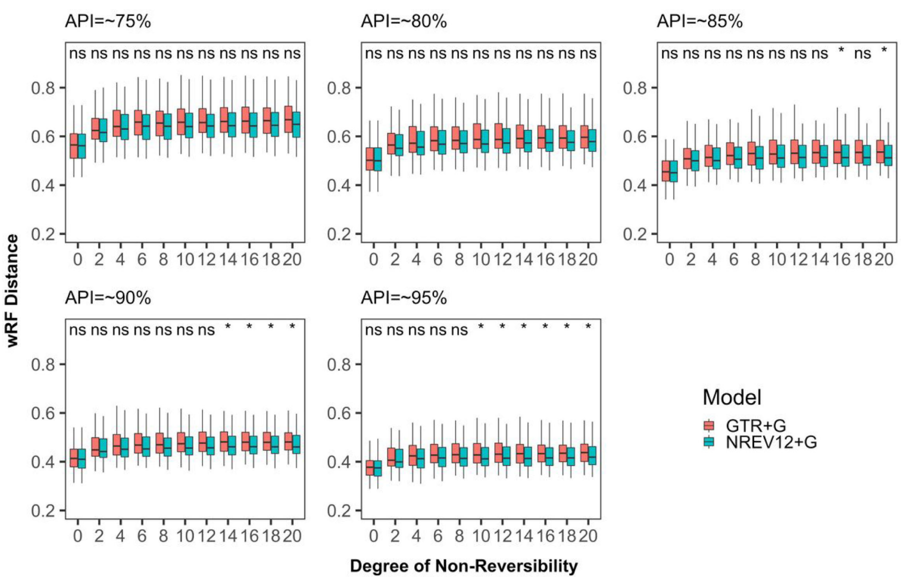

Figure 2

WeightedRobinson-Foulds distances between inferred and true phylogenetic trees for datasets simulated with different degrees of nucleotide substitution non-reversibility and different average pairwise sequence identities (APIs) (~75%%, ~80%, ~85%, ~90% and ~95%).

”ns” above a pair of box and whisker plots indicates a paired t-test adjusted p-value of greater than or equal to 0.05 and “*” indicate a paired t test adjusted p-value of <0.05

Table 1

AIC Scores and LRT results for double-stranded DNA virus datasets.

The lowest AIC scores indicating the best-itting models are in bold

Virus Family	Dataset	AIC Score GTR	AIC Score NREV-6	AIC Score NREV-12	P-Value GTR vs NREV-12	P-Value NREV-6 vs NREV-12	DNR
Papillomaviridae	APPV 6	**35099.5**	35108.0	35102.2	> 0.05	0.007	0.089
	HPV18_2	25202.9	**25174.6**	25179.2	< 0.001	> 0.05	0.323
	HPV45_2	23600.6	**23599.0**	23602.9	> 0.05	> 0.05	0.285
	HPV16_2	29734.0	**29664.5**	29665.4	< 0.001	> 0.05	0.371
	HPV31	24681.4	24677.3	**24672.8**	0.002	0.01	0.165
	HPV6_1	31199.1	31150.0	**31141.2**	< 0.001	< 0.001	0.451
	LPV	67165.7	67188.1	**67145.5**	< 0.001	< 0.001	0.42
	DPV	**69829.7**	69889.2	69835.1	> 0.05	< 0.001	0.056
	XPV	95455.6	95617.1	**95452.2**	< 0.001	< 0.001	0.072
	BATV	134821	134511	**133322**	< 0.001	< 0.001	0.402
Polyomaviridae	JC_2	**51806.7**	51819.6	51812.0	> 0.05	0.003	0.089
	BK_2	21472.6	21472.7	**21471.1**	0.03	0.03	0.244
	SV40	16859.8	**16858.0**	16858.4	0.037	> 0.05	0.567
	BPV	148614.9	**148573.8**	148585.2	< 0.001	> 0.05	0.064
Caulimoviridae	CMV	124083.9	124221.0	**123888.6**	< 0.001	< 0.001	0.351
	CSSV	145327.0	146575	**145202**	< 0.001	< 0.001	0.158
	SVBV	138575	138488.1	**138464.7**	< 0.001	< 0.001	0.174
	DBAV	**46495.5**	46514.1	46502.0	> 0.05	< 0.001	0.0335
	RTBV	**54987.9**	55350.1	54991	> 0.05	< 0.001	0.082
	BDV	376325.2	376647.6	**376029.9**	< 0.001	< 0.001	0.140
Siphoviridae	CLV	237362.3	237351.8	**237348.6**	**<** 0.001	< 0.01	0.070
Tectiviridae	TTIV	913864.9	913915.4	**913773.1**	< 0.001	< 0.001	0.279
Adenoviridae	FAV_C	3074086.7	3074207.5	**3073739.1**	< 0.001	< 0.001	0.169
	FAV_E	103482.3	103222.7	**102636.7**	< 0.001	< 0.001	0.357
	FAV_D	2326925.6	2325719.4	**2324784.5**	< 0.001	< 0.001	0.551
	FAV_A	705328.5	705436..5	**705197.8**	< 0.001	< 0.001	0.645
	HMAV_B	103796.7	103937.44	**103753.8**	< 0.001	< 0.001	10.890
	HMAV_D	1748635.2	1749769	**1748119.1**	< 0.001	< 0.001	0.646
	HMAV_C	2851144.5	2851357.1	**2851133**	0.006	< 0.001	0.0225
	HMAV_E	1915044.8	1915065.3	**1914998**	< 0.001	< 0.001	0.049

As NREV6 was not the best fitting model for most of the dsDNA virus datasets we infer that, in most dsDNA virus species, strand-specific substitution biases are not irrelevant. Further, the datasets where NREV6 was not the best fit are from species in families containing other species where NREV6 was the best fit, indicating that such strand-specific substitution biases are unlikely to be a consequence of some broadly conserved feature of viral life cycles in these families (such as, for example, ssDNA replicative intermediates). It is instead plausible that these differences may relate to:

1. differences in replication fidelity and/or proofreading efficiency on the leading and trailing DNA strands in some dsDNA virus species (Grigoriev, 1999): These differences are common in eukaryotes (Youri, Newlon, & KunkelThomas, 2002) (Furusawa, 2012) and prokaryotes (Fijalkowska, Jonczyk, Tkaczyk, Bialoskorska, & Schaaper, 1998) and, considering that the replication processes of dsDNA viruses analyzed here mirror those of their eukaryote hosts, it is perhaps unsurprising that most of these viruses also display some evidence of strand-specific substitution biases.

2. extra exposure to DNA damage of displaced template strands during unidirectional rolling circle replication in some papillomavirus species such as HPV16 could be a contributor to strand-specific nucleotide substitution biases (Kusumoto-Matsuo, Kanda, & Kukimoto, 2011).

3. extra time spent by non-coding strands in single-stranded dissociated states during RNA transcription in some papillomavirus and polyomavirus species (Fernandes & de Medeiros Fernandes, 2012): during transcription processes, the dissociated non-coding strand is transiently more exposed to damage than the coding strand (Wei, et al., 2010) which might also contribute to strand-specific substitution biases.

Similarly, and equally surprising, we found that NREV12 was overall the best supported model for dsRNA viruses (illustrated by the redder hues around the top corner of the dsRNA plot in Fig. 2).

NREV6 fit only two of the 31 dsRNA datasets better than both NREV12 and GTR (Human rotavirus A set H and Fiji virus). NREV12 was found to be the best fitting model for 21/31 datasets and GTR was the best fitting of 8/31 (Table 2). In all three Birnaviridae family datasets (which contains virus species with two genome segments) and in 17/22 of Reoviridae family datasets (which contain virus species with 10–12 genome segments) the NREV12 model provided the best fit. Based on the LRTs, strong overall support for NREV12 was found, with this model providing a significantly better fit ($p < 0.05$) in 27/31 dsRNA virus datasets relative to NREV6, and 23/31 datasets relative to GTR.

Table 2

AIC Scores and LRT results for double-stranded RNA datasets. The lowest AIC scores indicating the best fitting models are in bold.

Virus Family	Dataset	AIC score GTR	AIC score NREV-6	AIC score NREV-12	GTR vs NREV-12	NREV-6 vs NREV-12	DNR
Birnaviridae	AQBV	31754.9	31853.3	**31721.9**	< 0.001	< 0.001	0.219
	GBV_A	47176.9	47347.2	**47154.8**	< 0.001	< 0.001	0.142
	IPNV	79186.2	79221.9	**79182.4**	0.0145	< 0.001	0.162
	GBV_B	39313.7	39062.8	**38938.7**	< 0.001	< 0.001	0.201
Reoviridae	BTV_A	34803.5	34895.1	**34801.3**	0.03	< 0.001	0.042
	BTV_B	48849.9	48893.	**48837.1**	< 0.001	< 0.001	0.043
	BTV_C	28350.9	28386.5	**28350.8**	> 0.05	< 0.001	0.061
	BTV_D	24969.1	24947.3	**24894.0**	< 0.001	< 0.001	0.191
	BTV_F	20622.7	20708.5	**20610.2**	< 0.001	< 0.001	0.067
	BTV_G	63349.9	63485.0	**63345.9**	0.00426	< 0.001	0.040
	BTV_H	20596.7	20685.5	**20586.1**	< 0.001	< 0.001	0.118
	BTV_I	17592.7	17622.5	**17588.8**	0.01	< 0.001	0.095
	BRVA_C	41206.7	41187.4	**41137.1**	**<** 0.001	< 0.001	0.128
	HRVA_A	**17030.5**	17043.2	17035.5	> 0.05	0.003	0.036
	HRVA_B	**8275.1**	8280.3	8281.7	> 0.05	> 0.05	0.087
	HRVA_C	12815.1	12842.6	**12807.6**	0.003	< 0.001	0.132
	HRVA_D2	**8036.8**	8041.0	8043.7	> 0.05	> 0.05	0.057
	HRVA_E	**7045.9**	7056.1	7053.3	> 0.05	0.02	0.102
	HRVA_F	**7046.0**	7056.7	7053.4	> 0.05	0.02	0.0710
	HRVA_G	18424.2	18434.0	18425.1	> 0.05	< 0.001	0.123
	HRVA_H	20431.4	**20413.8**7	20420.5.6	0.002	> 0.05	0.163
	PRVA_A	28540.7	28441.9	**28398.7**	< 0.001	< 0.001	0.204
	PRVA_B	14757.7	14775.5	**14732.6**	< 0.001	< 0.001	0.351
	HRVC_A	6713.2	6718.2	**6712.3**	0.045	0.007	0.124
	PTOV	202011.3	202106.5	**201878.5**	< 0.001	< 0.001	0.039
	FJV_B	9274.1	**9250.0**	9250.9	< 0.001	> 0.05	0.194
Endornaviridae	EDV	1771992.8	1772689.1	**1771950.6**	< 0.001	< 0.001	0.121
	BPAV	**70386.5**	70540.2	70390.7	> 0.05	0.00	0.047
Totiviridae	TTV	617302.6	617462.6	**617172.9**	< 0.001	< 0.001	0.052
	GDV	**80435.8**	80396.5	80387.7	< 0.001	0.002	0.109

Virus Family	Dataset	AIC score GTR	AIC score NREV-6	AIC score NREV-12	GTR vs NREV-12	NREV-6 vs NREV-12	DNR
Hypoviridae	HPV	66859.8	66899.8	**66857.8**	0.03	< 0.001	0.057

We anticipated that NREV12 might fit many of these dsRNA datasets better than NREV6 simply because, during their infection cycles, the coding + strand of dsRNA viruses (the one from which protein translation occurs) tends to exist for longer periods within an infected cell than the non-coding –strand. Specifically, there are two main steps during double-stranded RNA virus replication (Wickner, 1993). Firstly, synthesis of the viral + strands from a dsRNA template occurs in the cytoplasm within viral particles. These + strands exist within the cell for prolonged periods in the absence of complementary -strands and are used as templates for translation of viral proteins. In the second step the + strands remaining after translation act as templates for -strand synthesis resulting in the formation of new dsRNA molecules. The + strands of dsRNA viruses are therefore likely more impacted by mutational processes, which in turn could explain the pervasive strand specific substitution biases seen in this group of viruses.

For the ssRNA and ssDNA viruses where one genome strand exists during the virus life cycle for far longer periods of time than the other such that complementary substitutions would not be expected to occur at similar rates, we anticipated that NREV12 should provide a better fit than both NREV6 and GTR. Indeed, for ssRNA viruses, NREV12 was a better fit than NREV6 and GTR for 40/47 of the ssRNA datasets and 24/33 of the ssDNA virus datasets (Fig. 2). Of the nine ssDNA virus datasets where NREV12 was not the best fitting model, GTR fit 7/9 better and NREV6 fit 2/9 better (Table 3). Of the seven ssRNA datasets (Table 4) where NREV12 was not the best fitting model, GTR fit 6/7 better and NREV6 fit 1/7 better.

Table 3

AIC Scores and LRT results for single-stranded DNA datasets.

The lowest AIC scores indicating the best fitting models are in bold.

Virus Family	Dataset	AIC Score GTR	AIC Score NREV-6	AIC Score NREV-12	P-Value GTR vs NREV-12	P-Value NREV-6 vs NREV-12	DNR
Nanoviridae	BBTV M	15044.3	15207.9	**14984.4**	< 0.001	< 0.001	0.662
	BBTV N	10605.6	10686.2	**10595.2**	< 0.001	< 0.001	0.533
	BBTV R	18484.5	18544	**18480.8**	> 0.05	< 0.001	0.609
	BBTV S	12718.9	12757.2	**12707.3**	< 0.001	< 0.001	0.728
	CCDV	**38622.7**	38632.0	38630.5	> 0.05	0.03	0.050
	MDV	36232.8	**36063**	36064	< 0.001	> 0.05	0.142
	PYDV	56138.4	56076.6	**56056.4**	< 0.001	< 0.001	0.187
	FBNS	100153.6	100135.6	**100120.5**	< 0.001	< 0.001	0.098
Geminiviridae	Begomo 5	**28192.1**	28311.9	28192.5	> 0.05	< 0.001	0.1995
	Begomo 6	16743.0	**16722.6**	16724.1	< 0.001	> 0.05	0.214
	Begomo 9	8517.6	8540.8	**8515.6**	0.03	< 0.001	0.312
	Dicot 1	44730.7	44594.3	**44583.3**	< 0.001	< 0.001	0.200
	Dicot 2	**39909.9**	39919.8	39917.9	> 0.05	< 0.001	0.100
	MSV	**252645.3**	254347.5	254347.5	< 0.001	< 0.001	0.144
	PanSV	94601.2	94600.3	**94593.7**	< 0.001	< 0.001	0.182
	WDV	35301.7	35313.2	**35253.8**	< 0.001	< 0.001	0.1033
Circoviridae	BFDV	17256.7	17262.7	**17246.7**	< 0.001	< 0.001	0.224
	DG_CV	**12754.8**	12779.5	12758.3	> 0.05	< 0.001	0.116
	PiCV	**19180.5**	19192.5	19191.0	> 0.05	0.04	0.117
	CCCC	84435.7	84377.4	**84315.3**	< 0.001	< 0.001	0.132
	BTC	262910.4	262060.1	**261985.4**	< 0.001	< 0.001	0.178
	POCV2	24940.9	24953.8	**24915.8**	< 0.001	< 0.001	0.162
	CCV	90307.9	90301.5	**90285.9**	< 0.001	< 0.001	0.114
Anelloviridae	TTV_1	825811	826800	**825292**	< 0.001	< 0.001	0.513
	TTSV	332287.9	332397.4	**332258.2**	< 0.001	< 0.001	1.560
Parvoviridae	MVM	26756.3	26743.9	**26686.9**	< 0.001	< 0.001	0.148
	HPV	67051.2	67080.1	**67001.8**	< 0.001	< 0.001	0.235
	CPV	85731	85695	**85689.3**	< 0.001	0.007	0.062
	PPV	163006.8	163090.7	**162995.9**	< 0.001	< 0.001	0.143
	CAV_P	37073.3	37115.5	**37065.7**	< 0.001	< 0.001	0.162
Microviridae	BMV	31175.3	31164.8	**31147.3**	< 0.001	< 0.001	0.188
Pleolipoviridae	APV	85700.2	85617.4	**85402.8**	< 0.001	< 0.001	0.204
	BPV	204797.5	204802.3	**204796.7**	0.04	0.007	0.064

Table 4

AIC Scores and LRT results for single-stranded RNA datasets.

The lowest AIC scores indicating the best fitting models are in bold.

Virus Family	Dataset	AIC Score GTR	AIC Score NREV-6	AIC Score NREV-12	P-Value GTR vs NREV-12	P-Value NREV-6 vs NREV-12	DNR
Astroviridae	HAV	94580.7	94926.3	**94548.1**	< 0.001	< 0.001	0.096
	BAV	188307.1	188572.9	**188144.9**	< 0.001	< 0.001	0.108
	MMV	281072.2	281094.5	**281076.9**	> 0.05	< 0.001	0.072
	PAV	150626.88	150827.6	**150609.5**	< 0.001	< 0.001	0.069
	CKV	90902.3	91233.1	**90873.0**	< 0.001	< 0.001	0.083
	GA	64998.5	65223.9	**64975.9**	< 0.001	< 0.001	0.110
	CAV_A	85558.8	85617.4	**85547.3**	< 0.01	< 0.001	0.076
Bromoviridae	CMV RNA1	34197.5	34198.8	**34147.7**	< 0.001	< 0.001	0.124
	CMV RNA2	**31398.2**	31455.9	31388.7	< 0.001	< 0.001	0.091
	CMV RNA3	**24337.2**	24360.3	24343.9	> 0.05	< 0.001	0.073
	AMS	**24337.2**	24360.3	24343.9	> 0.05	< 0.001	0.073
	PSV	67707	67786.5	**67691**	< 0.001	< 0.001	0.048
Caliciviridae	LAV	73042.8	73102.4	**72984.6**	< 0.001	< 0.001	0.120
	NoV	207667.2	207777.5	**207660**	< 0.001	< 0.001	0.047
	VSV	235936.4	236051.4	**235913.3**	< 0.001	< 0.001	0.046
Closteroviridae	CTV	**30062.2**	29980.4	29960.1	< 0.001	< 0.001	0.272
Flaviviridae	DGV_T1	69771.9	70030.5	**69776.2**	> 0.05	< 0.001	0.063
	JEV	146920.8	148101.5	**146885.5**	< 0.001	< 0.001	0.091
Hepeviridae	HPVE1	200439.5	200863.8	**200179.8**	< 0.001	< 0.001	0.073
	HPVE2	155709.1	155983.8	**155518.6**	< 0.001	< 0.001	0.088
Picornaviridae	ENV_A	552287.9	553535.5	**551794.1**	< 0.001	< 0.001	0.061
	HRV_A	102218.7	102267.0	**101550.7**	< 0.001	< 0.001	0.285
	AIV	101073.1	101136.7	**101052.2**	< 0.001	< 0.001	0.093
	AHP	139635.7	140119.6	**139506.9**	< 0.001	< 0.001	0.170
	ECV	82078.9	82181.0	**82065.8**	< 0.001	< 0.001	0.066
	CDV	130551.3	130896.7	**130478.3**	< 0.001	< 0.001	0.086
	TCV	53027.3	53029	**53023**	0.0151	0.0422	0.033
	FMDV	455180.6	455582.6	**454806.1**	< 0.001	< 0.001	0.117
Fusariviridae	FRV	**52413.1**	52470.6	52418.4	> 0.05	< 0.001	0.076

Virus Family	Dataset	AIC Score GTR	AIC Score NREV-6	AIC Score NREV-12	P-Value GTR vs NREV-12	P-Value NREV-6 vs NREV-12	DNR
Retroviridae	HIV1_setA	344014.4	344295.1	**343669.7**	< 0.001	< 0.001	0.237
	HIV1_M	80764.1	80829.5	**80668.1**	< 0.001	< 0.001	0.442
	HIV1_setC	180575.0	180702.3	**180494.4**	< 0.001	< 0.001	0.107
	HIV1_setD	298489.9	298695.3	**298260.6**	< 0.001	< 0.001	0.133
	HIV1_setE	289111.3	289292.1	**288941.9**	< 0.001	< 0.001	0.112
	HIV1_setF	214375.9	214692.2	**214289.4**	< 0.001	< 0.001	0.148
	EIV	126149	126365.4	**125300**	< 0.001	< 0.001	0.192
	BIV	**24505.2**	24506.9	24513.2	> 0.05	> 0.05	0.15
	FIV	164542.1	164487.9	**164260.4**	< 0.001	< 0.001	0.114
	CAV	351329.9	351871.5	**350721.9**	< 0.001	< 0.001	0.174
	SIV	110731.2	110816	**110663.3**	< 0.001	< 0.001	0.144
Filoviridae	Ebola_2	53147.3	**53143.0**	53149.9	> 0.05	> 0.50	0.264
Orthomyxo-viridae	Flu A 2	82872.8	83010.2	**82849.7**	< 0.001	< 0.001	0.27
	Flu B 1	50090.4	50144.1	**50060.9**	< 0.001	< 0.001	0.311
Coronaviridae	SARS-COV1	214715.3	214968.5	**214644.39**	< 0.001	< 0.001	0.198
	SARS-COV2	15715.4.2	15715.6	**15696.7**	< 0.001	< 0.001	1.536
	SARB	573966.3	573815.1	**572517.0**	< 0.001	< 0.001	0.301
	MERS-COV	516683.2	516983.4	**516608.9**	< 0.001	< 0.001	0.169

Based on the LRTs, strong overall support for NREV12 was found with this model providing a significantly better fit ($p < 0.05$) than NREV6 for 45/47 of the ssRNA virus datasets (Table 4) and 31/33 of the ssDNA virus datasets (Table 3). Similarly, based on LRTs, NREV12 provided a significantly better fit than GTR for 27/33 of the ssDNA virus datasets (Table 3) and 40/47 of the ssRNA virus datasets (Table 4).

We found that DNR estimates alone did not cleanly differentiate between datasets for which NREV12 was or was not best supported (Tables 1, 2, 3 and 4). For the 107 nucleotide sequence datasets with a model preference of NREV12, ten had estimated DNRs that were greater than 0.5, 13 had DNRs between 0.25 and 0.5 and 84 had DNRs between 0.0225 and 0.25. For the ten nucleotide sequence datasets with a model preference of NREV6, one had an estimated DNR greater than 0.5, four had estimated DNRs between 0.25 and 0.5 and five had estimated DNRs between 0.064 and 0.25 (Fig. 2). For the 24 nucleotide sequence datasets with a model preference of GTR, none had estimated DNRs greater than 0.5, one had an estimated DNR between 0.25 and 0.5 and the remainder had estimated DNRs between 0.0335 and 0.25.

The dsDNA virus dataset with the highest DNR was Human mastadenovirus D (DNR = 0.646), the dsRNA virus dataset with the highest estimated DNR was Porcine_rotavirus_B (0.351), the ssRNA virus dataset with the highest DNR was SARS-CoV-2 (DNR = 1.536) and the ssDNA virus dataset with the highest DNR was Torque teno sus virus (DNR = 1.56).

Therefore, while NREV12 appears to be generally more appropriate than either NREV6 or GTR for describing mutational processes in ssRNA, ssDNA, dsDNA, and dsRNA viruses, this might only be particularly relevant from a practical perspective when datasets of these viruses yield DNR estimates that are greater than 0.25. For such datasets NREV12 (and possibly NREV 6 in some instances) might be especially useful for both determining the direction of evolution across phylogenetic trees (i.e., it could potentially be used to root these trees) and for quantifying genomic strand-specific nucleotide substitution biases (Harkins et al., 2009).

Assessing the impacts of model misspecification on phylogenetic tree inference

To determine whether it might sometimes be worthwhile using NREV12 rather than GTR for phylogenetic inference when NREV12 is the best fitting nucleotide substitution model, we used simulated datasets to compare the accuracy of phylogenetic trees inferred using these models. Specifically, we simulated datasets with DNRs varying from 0 to 20 along known "true" phylogenetic trees (supplementary Figure S 1) with branches scaled to reflect branch tip sequences with APIs of ~ 75%, ~ 80%, ~ 85%, ~ 90% and ~ 95%. For each of five API levels, we therefore simulated 5500 datasets (comprising 100 datasets for each DNR = 0, 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20).

Phylogenetic trees were inferred from these 5500 simulated datasets and compared to the phylogenetic trees used to simulate the datasets (i.e., the true trees) using wRF distances to assess the impact of varying DNR on the accuracy of phylogenetic inference. We further tested whether the accuracy of phylogenetic inference could be improved for sequences that had evolved under DNR > 0 by using NREV12 instead of GTR. Specifically, for every simulated dataset a phylogenetic tree was inferred using GTR and another using NREV12 and the wRF distances of each of these trees to the true tree was determined. For each of the analysed DNRs a paired t-test (correlated t-test) was then used to compare the wRF scores of trees inferred using GTR and NREV12. We were particularly interested in determining whether trees inferred using a mis-specified model (i.e. GTR in this case) would be less accurate than trees inferred with a correctly specified model (i.e. NREV12).

We found that, irrespective of dataset diversity and the nucleotide substitution model used, phylogenetic inference tended to become more inaccurate (i.e. wRF scores increased) as DNR increased (Fig. 2). This tendency was, however, more pronounced when using a (miss-specified) GTR model than when using a (correctly specified) NREV12 model with, for any given dataset having DNR > 0, the use of NREV12 tending to yield more accurate phylogenetic trees than when GTR was used. There were, however, only statistically significant improvements ($p < 0.05$ paired t-test) in the accuracy of phylogenetic trees inferred using NREV12 relative to those inferred using GTR in lower diversity datasets (i.e. those with APIs of 85%, 90% and 95%); and then only for DNR > 8. It is noteworthy that the highest estimated DNR in any of the empirical datasets that we analysed was xxx: more than four-fold lower than the point where statistically significant differences in phylogenetic-inference accuracies became apparent in the simulated datasets.

Conclusion

The non-reversible nucleotide substitution model, NREV12, generally provides a substantially better fit to virus nucleotide sequence datasets than does the widely used reversible substitution model, GTR. NREV12 also generally provides a better fit to virus nucleotide sequence datasets than does NREV6; a non-reversible model that would be expected to best describe the evolution of double stranded genome sequences that display no strand-specific nucleotide substitution biases. This suggests that, contrary to our expectations, substantial strand-specific nucleotide substitution biases (i.e. estimated DNRs > 0.25) are common during viral evolution irrespective of genome type. Such biases should be expected for any viruses where one genome strand either is in existence for substantially longer periods of time than the other, or is more exposed to mutagenic processes than the other during transmission, replication, or gene expression.

We had anticipated that, given evidence of sequences evolving both non-reversibly and with strand-specific substitution biases, inferring trees using a model such as NREV12 that appropriately accounts for this might: (1) minimize the impact of increasing DNR on the accuracy of phylogenetic inference (i.e. wRF scores presented in blue in Fig. 2 might have been expected to not increase with increasing DNR); and (2) yield significantly more accurate phylogenetic inferences than when using GTR for all datasets where NREV12 was the most appropriate model and DNRs were greater than zero. However, increasing DNR clearly decreased the accuracy of phylogenetic inference even when using NREV12, and, for datasets where DNRs were greater than zero, using GTR did not consistently yield significantly less accurate phylogenetic inferences than those attained using NREV12. From a practical perspective, choosing a non-reversible nucleotide substitution model to construct phylogenetic trees from virus genome sequences that display strand specific nucleotide substitution biases is not guaranteed to yield more accurate phylogenetic trees. Nevertheless, in instances where strand specific substitution biases are higher than ~ 0.5 (such as are found in our SARS-CoV-2, Torque teno sus virus and Banana bunchy top virus datasets) it may be prudent to select a model such as NREV12 (such as is implemented in programs like IQTREE) over GTR as the better of two suboptimal choices.

The lack of available data regarding the proportions of viral life cycles during which genomes exist in single and double stranded states makes it difficult to rationally predict the situations where the use of models such as GTR, NREV6 and NREV12 might be most justified: particularly in light of the poor over-all performance of NREV6 and GTR relative to NREV12 with respect to describing mutational processes in viral genome sequence datasets. We therefore recommend case-by-case assessments of NREV12 vs NREV6 vs GTR model fit when deciding whether it is appropriate to consider the application of non-reversible models for phylogenetic inference and/or phylogenetic model-based analyses such as those intended to test for evidence of natural section or the existence of molecular clocks.

Abbreviations

- API:

  Average pairwise nucleotide sequence identity

- NREV6:

  6 Rate Non-Reversible Model

- NREV12:

  12 Rate Non-Reversible Model

- GTR:

  General time Reversible Model

- dsDNA:

  Double-stranded deoxyribonucleic acid

- dsRNA:

Double-strandedribonucleic acid

- ssDNA:

  Single-stranded deoxyribonucleic acid

- ssRNA:

  Single-strandedribonucleic acid

- DNR:

  Degrees of Non-Reversibility

- wRF:

  Weighted Robinson-Foulds distance

- LRT:

  likelihood ratio test

- AIC:

  Akaike information criterion test

Declarations

Ethics

The University of Cape Town ethics committee declared that this research did not need ethics approval due to the use of freely accessible nucleotide sequences obtained from the National Centre for Biotechnology Information Taxonomy database (http://www.ncbi.nlm.nih.gov/taxonomy) and the Los Alamos National Laboratory HIV sequence database (https://www.hiv.lanl.gov/content/index).

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on request.

Authors' contributions

RS contributed to the data collection, model testing, and interpretation of results and, together with PH, drafted the manuscript. DPM contributed to the conception and design of the study and interpretation of the findings. DPM and SLKP edited the HyPhy batch script used to conduct model tests and critically reviewed the manuscript. CBC, PH, CWM, FP, KR, SS contributed to the collection and model testing of viral genome sequences.

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Consent for Publication

Not applicable

Supplementary Information

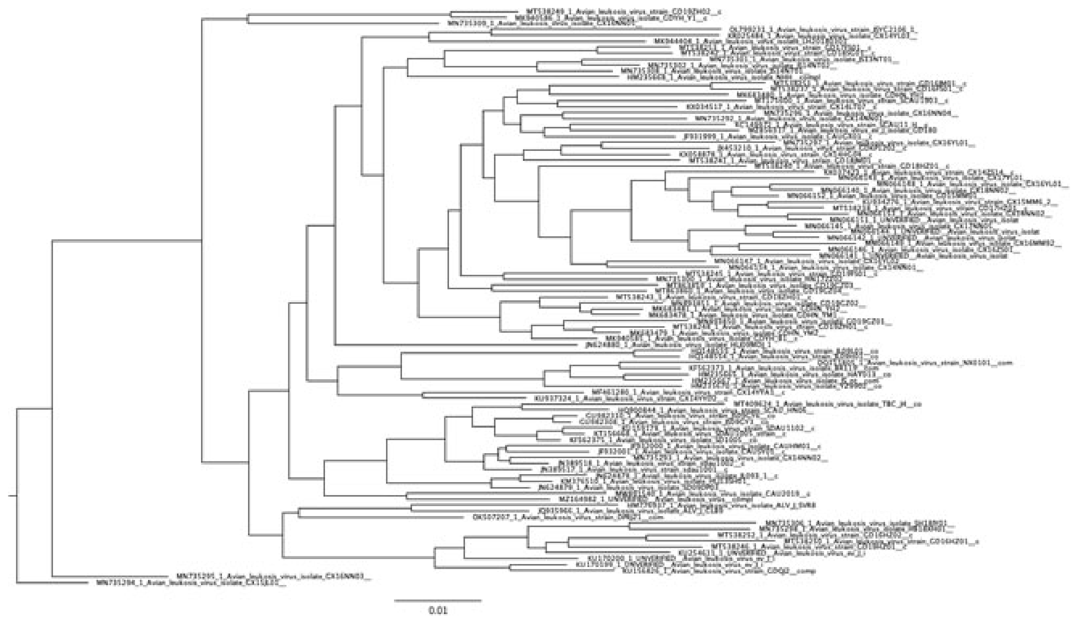

Figure S 1

Phylogenetic tree inferred from an alignment of real sequences (Avian Leukosis virus) that was used to simulate datasets with DNRs varying from 0 to 20.

The alignment of Avian Leukosis virus had an average sequence identity (API) of ~90% and the branches of this tree were scaled to produce four other trees reflecting branch tip sequences with approximate pairwise identities of ~75%, ~80%, ~85% and ~95%.

Table S1

Summary of the viral genome/genome component datasets used in the study.

Genome Type	Virus Family	Virus Genus	Virus Species	Dataset Name
ssDNA	Circoviridae	Circovirus	Beak and feather disease virus	BFDV
			Duck circovirus, Goose circovirus	DG_CV
			Columbine circovirus	PiCV
			Circovirus	CCCC
			Bat circovirus	BTC
			Porcine Circovirus 2	POCV2
			Cyclovirus	CCV
	Geminiviridae	Begomovirus	East Africa cassava mosaic virus, South African cassava mosaic virus	Begomo6
			Tomato yellow leaf curl virus	Begomo5
			Malvastrum yellow vein Yunnan virus, Cotton leaf curl Multan virus, Bhendi yellow vein India virus	Begomo9
			Tobacco yellow dwarf virus, Chickpea chlorosis virus, Chickpea yellows virus	Dicot_1
		Mastrevirus	Chickpea chlorotic dwarf virus	Dicot_2
			Maize streak virus	MSV
			Panicum streak virus	PanSV

			Wheat dwarf virus	WDV
	Anelloviridae	Annellovirus	Torque teno virus 1	TTV_1
			Torque teno sus virus	TTSV
	Parvoviridae	Aneptorquevirus	minute virus of mice, MVM	MVM
		Protoparvovirus	Human parvovirus	HPV
			Canine parvovirus	CPV
			porcine parvovirus	PPV
		Amdoparvovirus	Carnivore amdoparvovirus	CAV_P
	Nanoviridae	Babuvirus	Banana bunchy top virus	BBTV_M
				BBTV_N
				BBTV_R
				BBTV_S
		Nanovirus	Coconut foliar decay virus	CCDV
			Milk vetch dwarf virus full genome	MDV
			Pea necrotic yellow dwarf virus	PYDV
			Faba bean necrotic stunt virus	FBNS
	Microviridae	Microviruses	Microvirus	BMV
	Pleolipoviridae	Pleolipoviruses	Betapleolipovirus	BPV
			Alphapleolipovirus	APV
ssRNA	Astroviridae	Astroviruses	Human astrovirus	HAV
			Bovine astrovirus	BAV
			Mamastrovirus	MMV
			porcine astrovirus	PAV
			chicken astrovirus	CKV
			Goose astrovirus	GA

			Canine astrovirus	CAV_A
	Bromoviridae	Cucumovirus	Cucumber mosaic virus	CMV_RNA1
				CMV_RNA2
				CMV_RNA3
		Alphamovirus	Alphalfa mosaic virus	AMS
		Cucumovirus	Peanut stunt virus	PSV
	Caliciviridae	Lagovirus	Lagovirus	LAV
		coroviruses	Norovirus	NoV
		Vesivirus	Vesivirus	VSV
	Closteroviridae	Closterovirus	Citrus tristeza virus	CTV
	Flaviviridae	Flavivirus	Dengue virus	DGV_T1
			Japanese encephalitis virus	JEV
	Hepeviridae	Hepevirus	Hepatitis E virus	HPVE1
			Hepatitis E2 virus	HPVE2
	Picornaviridae	Enterovirus	Human Rhinovirus A	HRV_A
			Enterovirus A	ENV_A
		Teschovirus	Techovirus	TCV
		Aichivirus	Aichivirus	AiV
		Aphthovirus	Foot and mouth disease virus	FMDV
			Avihepatovirus	AHP
		Cardiovirus	Encephalomyo carditis virus	ECV
			Cardiovirus	CDV
	Fusariviridae	Fusarivirus	Fusariviruses	FRV
	Retroviridae	Lentivirus	Human immuno-deficiency virus 1	HIV1_setA
				HIV1_M
				HIV1_setC
				HIV1_setD

				HIV1_setE
				HIV1_setF
			Simian immuno-deficiency virus	SIV
			Bovine immunodeficiency	BIV
			Feline immunodeficiency	FIV
			equine infectious anemia virus	EIV
			caprine arthritis encephalitis virus	CAV
	Orthomyxo-viridae	Influenzavirus	Influenza virus A	FluA_2
			Influenza virus B	FluB_1
	Filoviridae	Ebolavirus	Ebola virus	Ebola_2
	Coronaviridae	Merbecovirus	Middle East respiratory syndrome	MERS-COV
		Sarbecoviruses	Severe acute respiratory syndrome coronavirus	SARS-CoV
			Severe acute respiratory syndrome coronavirus 2	SARS-CoV_2
			sarbecoviruses	SARB
dsDNA	Papillomaviridae	Alphapapillomavirus	Alphapapillomavirus 6	APPV 6
			Alphapapillomavirus 7	HPV18_2
				HPV45_2
			Alphapapillomavirus 9	HPV16_2
				HPV31
			Alphapapillomavirus 10	HPV6_1
			Bovine papillomavirus	BPV
			Lambdaapapillomavirus	LPV

			Deltapapillomavirus	DPV
			Xipapillomavirus	XPV
	Polyomaviridae	Polyomavirus	BK polyomavirus	BK_2
			JC polyomavirus	JC_2
			Bat polyomavirus	BPV
			Simian virus 40	SMV_40
	Caulimoviridae	Caulimovirus	cauliflower mosaic virus	CMV
			Cacao swollen shoot virus	CSSV
			Strawberry vein banding virus	SVBV
			Dioscorea bacilliform AL virus	DBAV
			Rice tungro bacilliform virus	RTBV
			badnavirus	BDV
	Siphoviridae	Escherichia virus Lambda	coliphage lambda	CLV
	Tectiviridae	Tectivirus	Tectivirus	TTIV
	Adenoviridae	Aviadenovirus	Fowl aviadenovirus C	FAV_C
			Fowl aviadenovirus E	FAV_E
			Fowl aviadenovirus A	FAV_A
			Fowl aviadenovirus D	FAV_D
		Mastadenovirus	Human mastadenovirus B	HMAV_B
			Human mastadenovirus D	HMAV_D
			Human mastadenovirus C	HMAV_C
			Human mastadenovirus E	HMAV_E
DsRNA	Birnaviridae	Avibirnavirus	Gumburo virus_setA	GBV_A
			Gumburo virus_setB	GBV_B
		Aquabirnavirus	Infectious pancreatic necrosis virus	IPNV
			Aquabirnavirus	AQBV

	Reoviridae	Orbivirus	Bluetongue_virus_setA	BTV_A
			Bluetongue_virus_setB	BTV_B
			Bluetongue_virus_setC	BTV_C
			Bluetongue_virus_setD	BTV_D
			Bluetongue_virus_setF	BTV_F
			Bluetongue_virus_setG	BTV_G
			Bluetongue_virus_setH	BTV_H
			Bluetongue_virus_setI	BTV_I
		Rotavirus	Bovine_rotavirus_A_setC	BRVA_C
			Human_rotavirus_A_setA	HRVA_A
			Human_rotavirus_A_setB	HRVA_B
			Human_rotavirus_A_setC	HRVA_C
			Human_rotavirus_A_setD2	HRVA_D2
			Human_rotavirus_A_setE	HRVA_E
			Human_rotavirus_A_setF	BRVA_F
			Human_rotavirus_A_setG	HRVA_G
			Human_rotavirus_A_setH	HRVA_H
			Porcine_rotavirus_A_setA	PRVA_A
			Porcine_rotavirus_A_setB	PRVA_B
			Human_rotavirus_C_setA	HRVC_A
		Orthoreovirus	Pteropine orthoreovirus	PTOV
		Fijivirus	Fijivirus_setB	FJV_B
	Totiviridae	Totivirus	Totivirus	TTV
		Giardiavirus	Giardiavirus	GDV
	Hypoviridae	Hypovirus	Hypovirus	HPV
	Endornaviridae	Endornavirus	Endornavirus	EDV
		Alphaendornavirus	Bell pepper alphaendornavirus	BPAV

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Reviewer #1 (Public Review):

The study by Sianga-Mete et al revisits the effects of substitution model selection on phylogenetics by comparing reversible and non-reversible DNA substitution models. This topic is not new, previous works already showed that non-reversible, and also covarion, substitution models can fit the real data better than the reversible substitution models commonly used in phylogenetics. In this regard, the results of the present study are not surprising. Specific comments are shown below.

Major comments

It is well known that non-reversible models can fit the real data better than the commonly used reversible substitution models, see for example,
https://academic.oup.com/sysbio/article/71/5/1110/6525257
https://onlinelibrary.wiley.com/doi/10.1111/jeb.14147?af=R
The manuscript indicates that the results (better fitting of non-reversible models compared to reversible models) are surprising but I do not think so, I think the results would be surprising if the reversible models provide a better fitting.
I think the introduction of the manuscript should be increased with more information about non-reversible models and the diverse previous studies that already evaluated them. Also I think the manuscript should indicate that the results are not surprising, or more clearly justify why they are surprising.

In the introduction and/or discussion I missed a discussion about the recent works on the influence of substitution model selection on phylogenetic tree reconstruction. Some works indicated that substitution model selection is not necessary for phylogenetic tree reconstruction,
https://academic.oup.com/mbe/article/37/7/2110/5810088
https://www.nature.com/articles/s41467-019-08822-w
https://academic.oup.com/mbe/article/35/9/2307/5040133
While others indicated that substitution model selection is recommended for phylogenetic tree reconstruction,
https://www.sciencedirect.com/science/article/pii/S0378111923001774
https://academic.oup.com/sysbio/article/53/2/278/1690801
https://academic.oup.com/mbe/article/33/1/255/2579471
The results of the present study seem to support this second view. I think this study could be improved by providing a discussion about this aspect, including the specific contribution of this study to that.

The real data was downloaded from Los Alamos HIV database. I am wondering if there were any criterion for selecting the sequences or if just all the sequences of the database for every studied virus category were analysed. Also, was any quality filter applied? How gaps and ambiguous nucleotides were considered? Notice that these aspects could affect the fitting of the models with the data.

How the non-reversible model and the data are compared considering the non-reversible substitution process? In particular, given an input MSA, how to know if the nucleotide substitution goes from state x to state y or from state y to state x in the real data if there is not a reference (i.e., wild type) sequence? All the sequences are mutants and one may not have a reference to identify the direction of the mutation, which is required for the non-reversible model. Maybe one could consider that the most abundant state is the wild type state but that may not be the case in reality. I think this is a main problem for the practical application of non-reversible substitution models in phylogenetics.

Reviewer #2 (Public Review):

The authors evaluate whether non time reversible models fit better data presenting strand-specific substitution biases than time reversible models. Specifically, the authors consider what they call NREV6 and NREV12 as candidate non time-reversible models. On the one hand, they show that AIC tends to select NREV12 more often than GTR on real virus data sets. On the other hand, they show using simulated data that NREV12 leads to inferred trees that are closer to the true generating tree when the data incorporates a certain degree of non time-reversibility. Based on these two experimental results, the authors conclude that "We show that non-reversible models such as NREV12 should be evaluated during the model selection phase of phylogenetic analyses involving viral genomic sequences". This is a valuable finding, and I agree that this is potentially good practice. However, I miss an experiment that links the two findings to support the conclusion: in particular, an experiment that solves the following question: does the best-fit model also lead to better tree topologies?

On simulated data, the significance of the difference between GTR and NREV12 inferences is evaluated using a paired t test. I miss a rationale or a reference to support that a paired t test is suitable to measure the significance of the differences of the wRF distance. Also, the results show that on average NREV12 performs better than GTR, but a pairwise comparison would be more informative: for how many sequence alignments does NREV12 perform better than GTR?